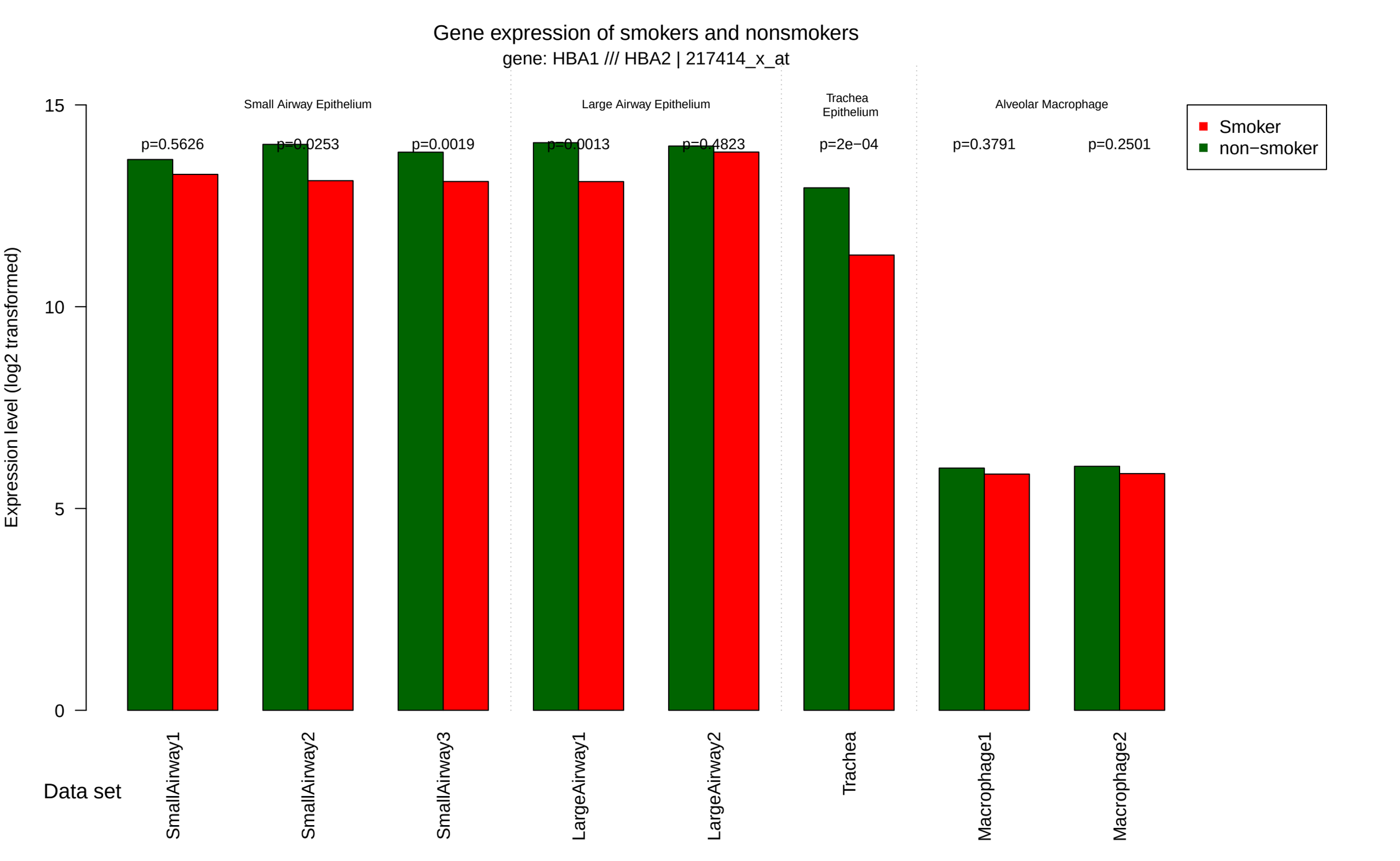
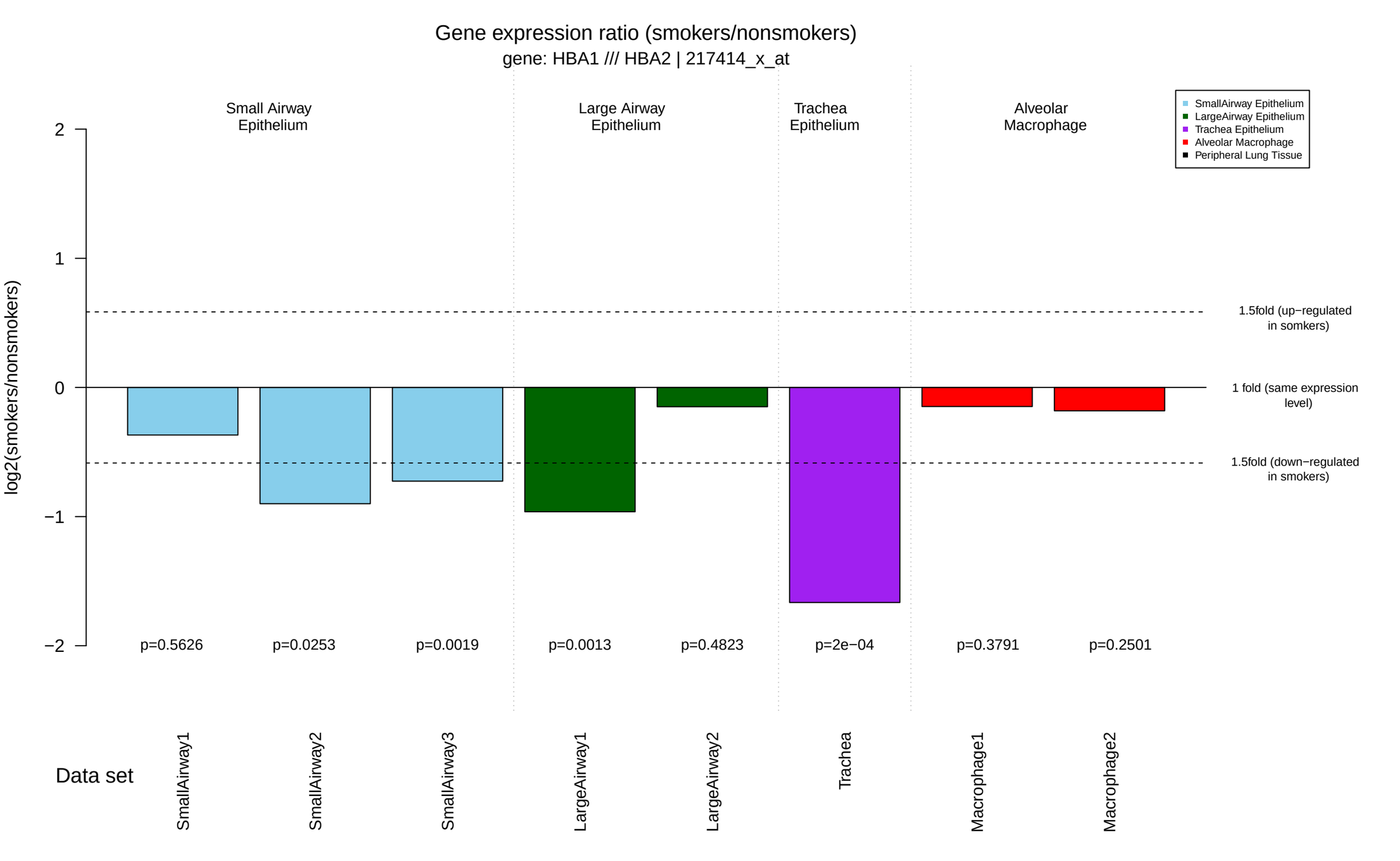
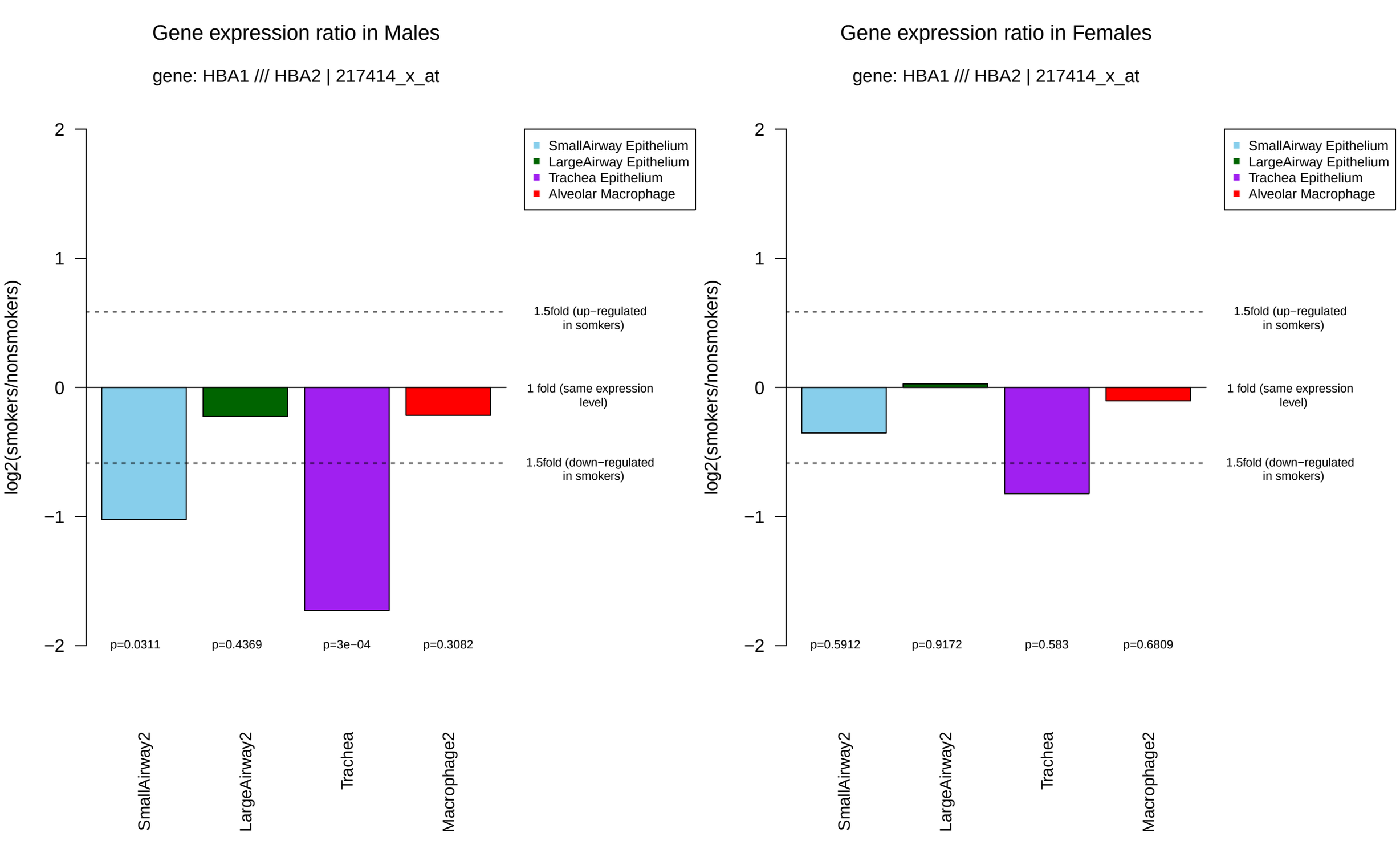
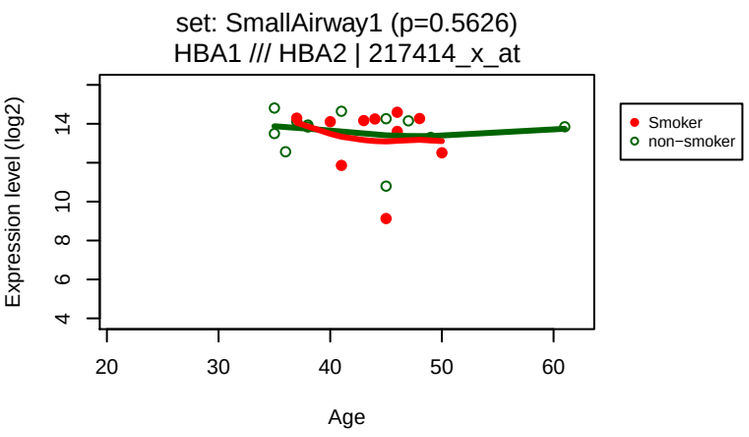
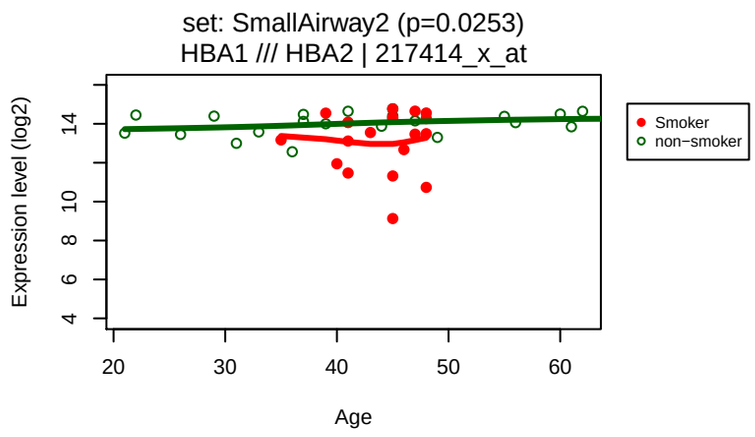
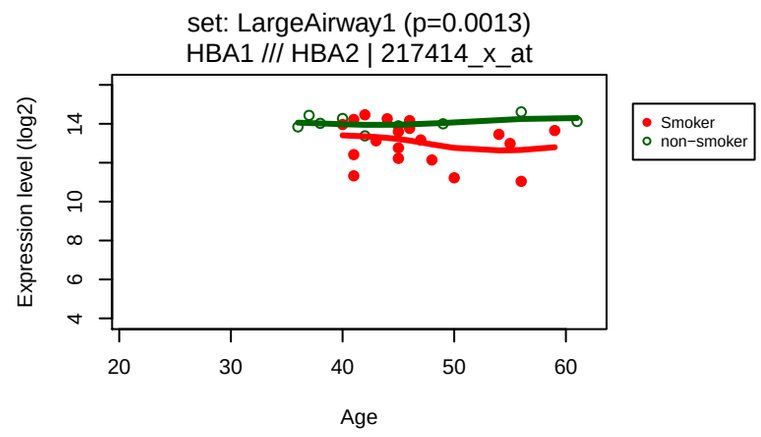

Gene expression of smokers and nonsmokers
gene: HBA1 /// HBA2 | 217414_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5626
p=0.0253
p=0.0019
p=0.0013
p=0.4823
p=2e−04
p=0.3791
p=0.2501
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

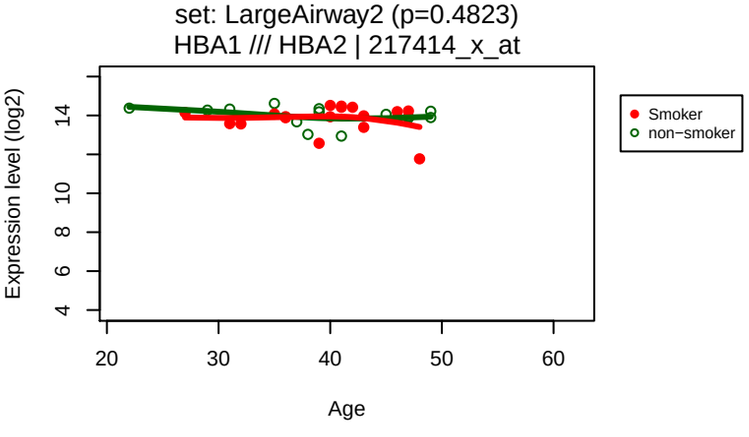

Gene expression ratio (smokers/nonsmokers)
gene: HBA1 /// HBA2 | 217414_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5626
p=0.0253
p=0.0019
p=0.0013
p=0.4823
p=2e−04
p=0.3791
p=0.2501
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

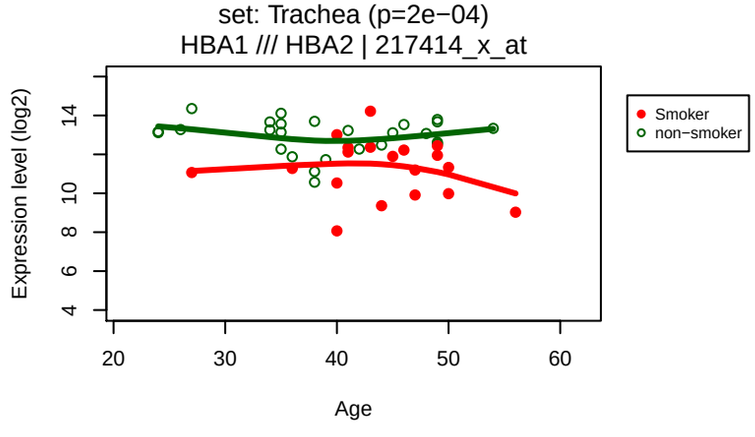

Gene expression ratio in Males
gene: HBA1 /// HBA2 | 217414_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0311
p=0.4369
p=3e−04
p=0.3082
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBA1 /// HBA2 | 217414_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5912
p=0.9172
p=0.583
p=0.6809
SmallAirway2
LargeAirway2
Trachea
Macrophage2

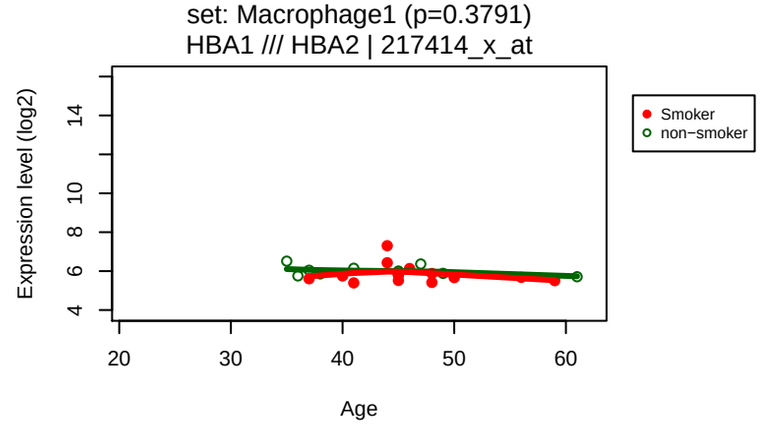

set: SmallAirway1 (p=0.5626)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker

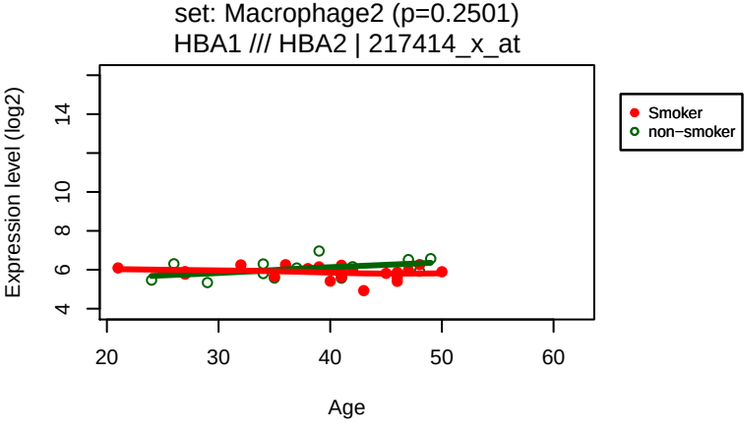

set: SmallAirway2 (p=0.0253)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker

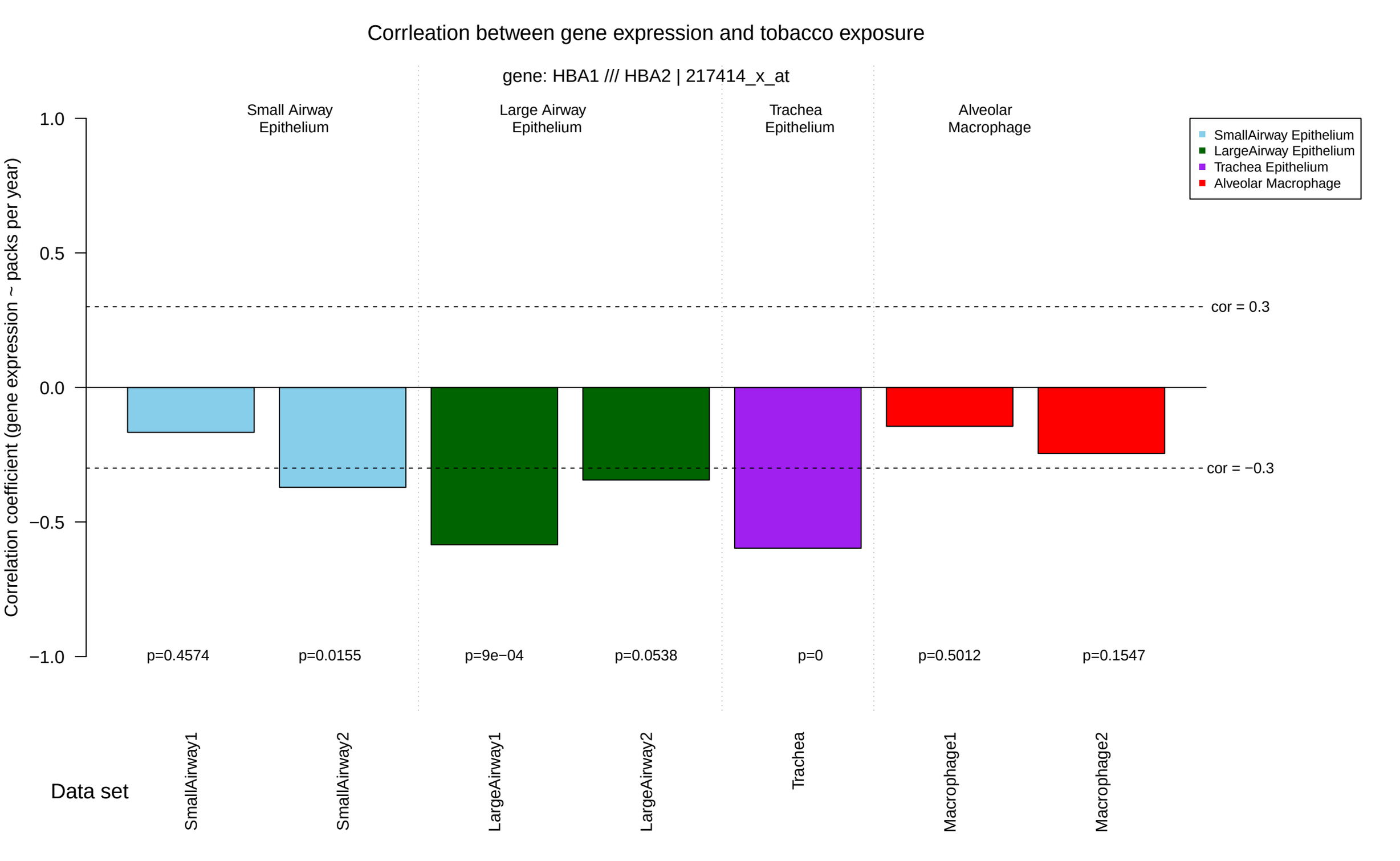

set: LargeAirway1 (p=0.0013)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4823)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3791)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2501)
HBA1 /// HBA2 | 217414_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBA1 /// HBA2 | 217414_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4574
p=0.0155
p=9e−04
p=0.0538
p=0
p=0.5012
p=0.1547
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2